Data Structures - Unit Wise 5M and 10M Important Questions

UNIT I

10 Marks

1. Explain Structures in C with suitable examples.
2. Explain Nested Structures and Arrays of Structures with examples.
3. Explain Self-Referential Structures and their applications.
4. Compare Structures and Unions with memory representation.
5. Explain Enumerated Data Types (enum), typedef, and Bit Fields with examples.
6. Discuss the classification of Data Structures with examples.

5 Marks

1. Define Structure and explain its syntax.
2. Explain Arrays of Structures.
3. Explain Structures and Functions.
4. What is a Self-Referential Structure?
5. Differentiate between Structure and Union.
6. Explain enum with an example.
7. Explain typedef with an example.
8. What are Bit Fields?
9. Define Data Structure.
10. Differentiate Linear and Non-linear Data Structures.

UNIT II

10 Marks

1. Explain Abstract Data Types (ADT) and their implementation.
2. Explain Time and Space Complexity analysis.
3. Explain Quick Sort with algorithm and example.
4. Explain Merge Sort with algorithm and example.
5. Explain Radix Sort with algorithm and example.
6. Compare Quick Sort, Merge Sort, and Radix Sort.

5 Marks

1. Define Linear Data Structure.
2. What is an ADT?
3. Explain Time Complexity.
4. Explain Space Complexity.
5. Write the Quick Sort algorithm.
6. Write the Merge Sort algorithm.
7. Explain the working of Radix Sort.
8. Compare Quick Sort and Merge Sort.
9. List advantages of Merge Sort.
10. List applications of Sorting.

UNIT III

10 Marks

1. Explain Singly Linked List representation and operations.
2. Explain insertion and deletion operations in Singly Linked Lists.
3. Explain Doubly Linked Lists and their operations.
4. Explain Circular Linked Lists and their operations.
5. Compare Arrays and Linked Lists.
6. Discuss applications of Linked Lists.

5 Marks

1. Define Linked List.
2. Explain node representation in a Linked List.
3. Explain insertion at the beginning of a Singly Linked List.
4. Explain deletion in a Singly Linked List.
5. Explain Doubly Linked List.
6. Explain Circular Linked List.
7. Differentiate Singly and Doubly Linked Lists.
8. Differentiate Arrays and Linked Lists.
9. List applications of Linked Lists.
10. Explain traversal operation in a Linked List.

UNIT IV

10 Marks

1. Explain Stack operations with algorithms.
2. Implement Stack using Arrays.
3. Implement Stack using Linked Lists.
4. Discuss applications of Stacks.
5. Explain Queue operations with algorithms.
6. Implement Queue using Arrays.
7. Implement Queue using Linked Lists.
8. Explain different types of Queues.
9. Explain Circular Queue with suitable example.
10. Discuss applications of Queues.

5 Marks

1. Define Stack and its properties.
2. Explain Push and Pop operations.
3. Explain Stack implementation using Arrays.
4. Explain Stack implementation using Linked Lists.
5. List applications of Stacks.
6. Define Queue and its properties.
7. Explain Enqueue and Dequeue operations.
8. Explain Circular Queue.
9. Explain Priority Queue.
10. Differentiate Stack and Queue.

UNIT V

10 Marks

1. Explain Tree Terminology with examples.
2. Explain different types of Trees.
3. Explain Binary Tree Traversals with examples.
4. Explain Binary Search Tree insertion and deletion operations.
5. Compare Binary Tree and Binary Search Tree.
6. Explain Graph representation using Adjacency Matrix and Adjacency List.
7. Explain Breadth First Traversal (BFS) with example.
8. Explain Depth First Traversal (DFS) with example.
9. Compare BFS and DFS.
10. Discuss graph terminology and representations.

5 Marks

1. Define Tree and its terminology.
2. Explain Binary Tree.
3. Explain Binary Search Tree.
4. Explain Preorder Traversal.
5. Explain Inorder Traversal.
6. Explain Postorder Traversal.
7. Define Graph and its terminology.
8. Explain Adjacency Matrix.
9. Explain Adjacency List.
10. Differentiate BFS and DFS.